Process Performance Measurement System Characteristics: An Empirically Validated Framework

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Abstract. With the proliferation of business process reforms in organizations, the need for process-centric performance measurement systems is discussed in several studies. However, although there is considerable research on Performance Measurement Systems (PMSs) in general business contexts, research on performance measurement in process-centric contexts (e.g. within the BPM field) is scarce, and the characteristics of such process performance measurement systems (PPMSs) still remain poorly conceptualized, hindering their design and implementation. PPMSs are different from traditional performance measurement systems as the data gathered, and the information generated and disseminated are focused on processes, rather than on functions. This paper presents a PPMS characteristics framework, resulting from a multi-staged study design. Initially 38 PPMS characteristics were identified through a structured literature review which were then re-specified and confirmed with two in-depth case studies. The study findings resulted in an empirically supported PPMS characteristics framework, consisting of 30 PPMS characteristics grouped within 7 core themes, namely; (1) Quality characteristics; (2) Measurement Scope; (3) Contextual features considered in designing PPMS; (4) Relationship to organizational systems and structures; (5) Efficiency of information gathering and use; (6) Feedback and reporting; and (7) Potential uses of feedback generated. This is the first evidence-based synthesis of PPMSs characteristics and it provides a clear conceptualization and an understanding of what aspects a PPMS should comprise. The resulting framework will assist practitioners in designing and redesigning measurement systems in process centric contexts, which could in turn better support processes such as identifying improvement needs, measuring improved processes, benchmarking and controlling the processes, process maturity assessments, and benefits realization. The findings will also facilitate future research on PPMSs.

Keywords: Process performance measurement · Performance measurement · Business process management · Characteristics · Case study research

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1 Introduction

Performance measurement (PM) has been recognized as significant in many fields and process performance measurement is a well-established critical success factor for business process management (BPM) [1–3]. A process performance measurement system (PPMS) is expected to provide a systematic measurement of business processes [4] and it is a system that gathers process performance relevant data; compares the performance related values with other historical and target results/values and disseminates the results to the relevant process actors [5].

As stated by Tregear [6] "*if you don't measure process performance, you can't do process management - and you won't know if you are doing process improvement*" (p. 5). The 'measurement' aspect of process performance has therefore gained attention both in BPM practice [7] and as a significant and growing research area [8, 9]. It is important for the measures to be 'process-centric' in order to mitigate the risks faced in achieving the expected results of the process changes [10].

Substantial benefits can be gained in strategic and operational management through the use of effective process performance measures [11]. PPMSs provide evidence for the value/impact of efficient and effective processes, enabling these improved processes to be better embedded in daily operations [12]. The BPM literature recognizes the importance of process-centric performance measurements to enable better BPM adoption outcomes [4, 12]. However, organizations tend to adopt different elements of PPMSs, and measure processes '*ad-hoc*' and not continuously [13, p. 615], and the measurement of the impact of process improvements is often lacking [e.g. 4, 5, 8, 9]. Moreover, process improvements are more likely to have a positive impact on overall organizational performance when the performance measurement system (PMS) is designed or redesigned and deployed appropriately [14]. Likewise, inadequate measurement generates uncertainties regarding the impact of process improvement initiatives [15]. Thus, a well-designed process performance measurement system is needed for planning, designing and controlling business processes from the early stages of the process change/improvement initiatives [16, 17].

Traditional performance measures that focus on individual organizational functions do not cater well to contemporary business needs [18, 19] and the measurement needs of process improvement initiatives [20]. How to construct such process focused performance measures that do cater to the measurement needs of process improvement initiatives is not yet well understood. While there is considerable research on PMSs in general business contexts, research on performance measurement in process-centric contexts (e.g. within the BPM field) is scarce. For any field to develop and be more relevant to theory and practice, the characteristics of a system under consideration need to be made specific and explicit [21]. Explication of characteristics also facilitates consistent measurement and effective comparisons with other PMSs [22]. Thus, to develop and deploy relevant PPMSs or to redesign existing PMSs to better suit process-centric contexts, it is essential to identify the required characteristics of a PPMS; but such is not available in the current body of knowledge.

The research question driving this study is: ***What are the characteristics of a process performance measurement system?*** The aim of this work is to derive a comprehensive list of PPMS characteristics which are essential to consider when designing and deploying a PPMS. The framework with its evidence supported structured set of characteristics, can serve as a useful reference point for PPMS practice and future research. In the context of this study, 'characteristics' are the key features (or attributes) that form and describe a PPMS (i.e. the 'what' aspects). Note that we do not consider the PPMS design-steps (the 'how' aspects) when aiming to characterize PPMSs. While the PPMS characteristics could have specific design implications across the diverse stages of developing a PPMS, the focus of this paper is solely on deriving the PPMS characteristics (the design guidelines will be part of future research that follows).

A critical synthesis of the existing literature on PPMS characteristics is presented next (Sect. 2), to further justify the motivation for this work. The study comprised a multi-staged design as outlined in Sect. 3, with a structured literature review (SLR) (Sect. 3.1) followed by two in-depth case studies (Sect. 3.2). The study findings resulted in an empirically supported PPMS characteristics framework, consisting of 30 PPMS characteristics grouped within 7 core themes (Sect. 4).

2 Overview of the Existing Literature on PPMS Characteristics

A review of the literature on performance measurements in process-centric contexts confirms the limited research on PPMS characteristics. However, considerable research relating to other aspects of PMSs in process-centric contexts does exist. For instance, many scholars discuss the need for the adaptation of the traditional PMSs to BPM contexts [5, 6, 9, 12, 23, 24] including the need for methods to depict the financial/economic consequences of the process improvement efforts [17, 25, 26] as well as the importance of measuring process orientation [27]. Some refer to *managerial issues*, such as the role of process performance measurement in BPM adoption outcomes [4]; example cases of benefits obtained by tracking process performance [28]; the value of PPMSs [29]; factors influencing the ineffectiveness of PPMSs [29]; process performance measurement as a component of maturity assessment [30, 31]; process owner role and process performance measurement [32]; having a strategic approach within the organization for BPM and its impact on process performance measurement [11]. Others present certain *PPMS attributes* - defining process performance indicators [33]; review on the current state of research on PPMSs [34] and specific PPMS indicators and metrics [35].

Even though the need for a clearer conceptualization of PPMSs is often mentioned, detailed discussions on PPMS characteristics are still very limited. A systematic literature search (see Sect. 3) conducted on PPMSs revealed only eight papers that proposed potential characteristics for process-centric contexts. Three papers concerned PPMS characteristics in general and the other five discussed performance measurement characteristics in specific process improvement contexts; such as Just-in-Time (JIT) and Total Quality Management (TQM) environments.

With regard to the three papers focusing specifically on PPMS characteristics, Kueng and Krahn [24, p. 153] mention only one characteristic. i.e. "*PPMS should* *present an integral and holistic view of the performance of business processes*". Kueng [5] suggests three characteristics for a PPMS. It should be: focused on processes; evaluate performance holistically; and designate responsible parties (to each indicator to be measured). As suggested by Wieland et al. [36], this paper also argues that these characteristics are incomplete and too vague to operationalize. Wieland et al. [36] discuss PPMS characteristics essential for successful customer-orientation and identified customer demands and design features of PPMSs through a literature review. The limitations of the Wieland et al. [36] framework are; the 'customer only' focus, and how they mix the characteristics and design-steps to be considered when designing a PPMS etc. (as presenting them all together does not assist in the delineation of the precise characteristics of a fully developed PPMS).

The other five papers discussed PMS but did not focus on PPMS explicitly. Two papers discussed the characteristics of a PMS in a Just-in-Time environment [37, 38]. Schalkwyk [39] discussed what the PMS should consist of, within a TQM environment. The use of the PMS framework to enable and guide sound process re-engineering endeavours in maintenance was presented by Kutucuoglu et al. [40]. Bond [41] discussed PMS within a kaizen and re-engineering context and proposed that the characteristics of the different stages of process life cycles need to be considered when determining the performance metrics and when designing approaches for monitoring and controlling the processes. All the characteristics proposed in these eight papers are considered in our analysis and included within the preliminary set of 38 characteristics.

It was evident from the review that the PPMS literature to date is lacking a comprehensive synthesis. We also noted that the existing discourses often only addressed specific process-centric aspects and did not cover some of the essential characteristics that should be found in any performance measurement system (e.g. controls to minimize the opportunity for manipulation of the measures and results etc.). With the underlying assumption that the PPMS is a type of PMS, we argue that for a characterization of PPMS to be complete, it should not only focus on process centric specificities, but also have all essential features that any PMS should have. We maintained this view in our research design and the development of our framework.

3 Methodology

This study followed a multi-phased approach, using a structured literature review to identify potential PPMS characteristics and two in-depth case studies to provide empirical validation. The phases are summarized in Sects. 3.1 and 3.2.

3.1 A Structured Literature Review (SLR) to Synthesize PPMS Characteristics

The authors aimed to collate and synthesize the scattered knowledge on the characteristics of PPMSs (for the specific process-centric features) and characteristics of PMSs in literature (based on the assumption that PPMS is a type of PMS and hence should include the basic features of a PMS adapted to a process-centric context). The SLR applied in this study followed the guidelines proposed by Bandara et al. [42].

After a background search on the topic areas, the main keywords - 'performance measurement system' and 'process performance measurement system' were selected. A search string combining these two terms was used to search in ABI/INFORM, EBSCOhost and Emerald Insight in August 2017, resulting in 665 potentially relevant papers. A relevance check was carried out by two of the authors at several iterations with the primary rule being; 'the paper must *directly* discuss characteristics of PMSs or PPMSs'. Conceptual papers (i.e. those without supporting empirical evidence) were included for completeness purposes. 38 papers resulted from this relevancy screening. Most of the papers were removed from the analysis as they either discussed other aspects of PMS/PPMS such as measurement system design stages or did not discuss the characteristics in a collective manner. Backward and forward searches were carried out to identify further relevant papers, and 15 peer-reviewed papers and 3 book chapters were added. The 3 book chapters were not peer-reviewed but were included as many other papers referred to them and it is recognized that this sort of 'grey literature' can add value, especially to under-researched areas [43].

The 56 sources[1] were then exposed to a detailed coding protocol- with coding guidelines [44] that were specifically designed and pre-tested. PMS/PPMS characteristics were inductively extracted and recorded using an Excel Spreadsheet [42], resulting in a total of 418 open codes. These were tabulated separately as PMSs and PPMSs characteristics. Iterative reviews by the research team of these open codes resulted in some of the characteristics being broken down into further elements (to represent atomic sub-themes) and some (those that were very similar) being merged together, resulting in 473 total re-specified open codes. In the next round, axial coding was applied by forming coding families (following [45, 46]); resulting in the open codes being synthesized into 38 discrete PPMS characteristics (C1–C38). Appendix A lists and describes the 38 PPMS characteristics, and shows the sources supporting each. These characteristics were validated through two case studies as described in the next section.

3.2 In-depth Case Studies to Validate a-Priori PPMS Characteristics

Multiple-case designs are desirable when the intent of the research is description, theory building, or theory testing [47] and multiple case studies enable researchers to explore differences within and between cases and to replicate the findings across cases [48]. Thus, two case studies are conducted to validate the PPMS characteristics instead of a single case study.

The cases were selected based on a pre-defined case selection criterion (recommended in other studies (i.e. [49, 50])). A qualifying case had to be an improvement initiative implemented in a core business process in an organization that had implemented multiple process improvement initiatives and have a PPMS and/or PMS in place within the organization). Two cases in two large scale multinational manufacturers in Sri Lanka were studied. One was an apparel manufacturer (subsequently

[1] See References provided in Appendix A.

referred to as AMC) and the other was a tyre manufacturer (subsequently referred to as TCC). The unit of analysis was the process improvement initiative).

In AMC, the process improvement initiative concerned an improvement to one of their core business processes – 'sewing process' was studied in depth. AMC is the first apparel manufacturer in Sri Lanka, to introduce this kind of a process improvement initiative (subsequently referred to as the 'Dancing module'). They planned and tested it for one and a half years. Implementation was carried out in stages, where firstly 60 team members who worked in 2 lines were divided into 3 lines of 20 members each to increase productivity. Thereafter all the lines in the plant were converted. The team members were the lowest level employees in the plant and prior to the dancing module, they were always seated when sewing the garments. With the change, they stand and stitch the whole day and rotate to at least 3 machines. Most of the team members were only exposed to the seated mode of working, in this 30-year-old manufacturing plant. Thus, introducing the dancing module was a major transformation.

In TCC, the unit of analysis studied was an improvement to the mould operation (subsequently referred to as 'change in mould operation') - one of the core and critical processes in their tyre manufacturing process. TCC is the largest multinational tyre manufacturer in Sri Lanka with eleven plants. This was a unique case, in the following regard; that the initiative took place after a set of failed initiatives introduced by the Centre of Excellence for BPM in the organization, together with external consultants. According to the study respondents, the main reason for the failure in the earlier initiatives had been the lack of properly aligned measures. They planned and tested the new change initiative between 6–12 months in one plant prior to implementing to the full factory floor, which was implemented next within three plants and later expanded to the other eight plants.

Interviews and document analysis were used as the primary data collection mechanisms. Semi-structured interview protocols were used to guide the interview process which was conducted in two stages; for primary data collection, and for results confirmation. 12 interviews were conducted at AMC (*with - Manager Lean Enterprise - BPM Centre of excellence, Deputy General Manager – HR, Group Head of Finance, Management Accountant, Operations Manager, Production Control Unit Manager, Head/Manager Lean Enterprise, Project Champion cum Industrial Engineering Executive, Lean Implementation Executive and three Sewing Machine Operators*) and 10 interviews at TCC; *with - Program Manager (Strategic Supply Chain Process Improvements), Plant Director, Industrial Engineering Senior Manager, Product and Process Senior Engineer, Product and Process, IE & Lean Senior Executive Engineer and two Shift in-charges*). The duration of interviews varied between 30–90 min, and some respondents were approached twice for further clarifications. See Table B.1 of Appendix B for further details. NVivo tool was used as a data/evidence management tool for managing and coding the interview transcripts. A coding rule book and a pre-defined node structure were used for the analysis. For instance, a statement such as: "*The measures will be defined in detail very clearly so that we know what we are measuring and how and for what objective they will be introduced*" is coded under C1-Performance measures being clearly defined, with an explicit purpose. See Appendix B, for detailed case evidence pertaining to the 38 characteristics. A summary of case study results is presented next.

4 Summary Case Study Results

The a-priori 38 PPMS characteristics derived from the literature review (see Sect. 3.1 and Appendix A) were re-specified and confirmed through two case studies and adjusted (based on case evidence) to derive the final PPMS characteristics framework. Two new PPMS characteristics were identified and added to the list of characteristics (Sect. 4.1), and some characteristics with weak supporting evidence were still taken forward (Sect. 4.2), and some characteristics were merged due to clear similarities observed through the case data (Sect. 4.3). They were then categorized into the seven core themes as discussed in Fig. 1 in Sect. 5.

4.1 Newly Added Characteristics

Two characteristics that were not covered in the initial 38 characteristics (see C1–C38 Appendix A) were added and are explained further below.

(New-1) - Performance measures should provide the ability to challenge the jobs and encourage employees to work at a higher scale. [This was taken as a characteristic due to statements such as: *"our PMS is always challenging the job roles. We try to make the job challenging to employees so that they will stretch towards higher targets. That is an expectation from the employees too"* - Production Control Unit Manager (AMC) and "*in the first six months we didn't have proper measurements/KPIs, it was only basic things like monitoring within 24 h. We tightened the KPIs step by step ... They did the same work, but the measures were getting tight and they had to work at a higher speed*" - Industrial Engineering Manager (TCC)].

(New-2) - Performance measurements should be able to guide the employees on how to perform the expected tasks and to what extent they should be performed. [This was taken as a characteristic due to statements such as: "*people should know the guideline to perform. More than pushing the people, KPIs are like guidelines for them. When you take a job description it is very broad. It doesn't show the KPIs. This is important. When you work you should know what the goals are. This is guidelines for them to work and also for us to have as a rewards base*" - Deputy General Manager - HR (AMC) and "*together we define the KPIs and they get the index cards and they will make their staff work on them. Then they know where they should reach and what they need to reach there*" - Plant Director (TCC)].

4.2 A-Priori Characteristics with Conflicting Empirical Support

While the a-priori 38 characteristics were instantiated across the two cases (see Appendix B for supporting case evidence), two (C6 and C15) were subject to conflicting views and were only weakly confirmed; but were still included in the final list.

C6 (The set of performance measures should be few but complete and critical). C6 was not instantiated with evidence from current practice by both cases. But, AMC respondents believed that it is important to have few measures and they were attempting to reduce the number of measures and KPIs. TCC respondents did not believe in having few measures but did believe in having the most important set of

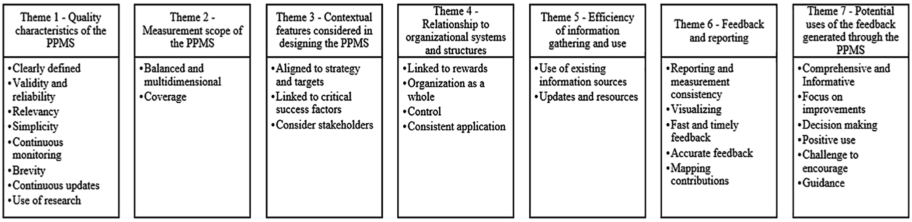

Fig. 1. The resulting PPMS Characteristics framework.

measures to cover all important aspects that need to be measured for decision making related to the process improvement initiatives. With C15 (Linked to critical success factors and key business drivers) the respondents of both cases saw value in this characteristic but stated that it is not consistently required for all process improvement initiatives. Critical success factors (CSFs) are identified and measured only when a process improvement initiative is costly and/or complex. Some CSFs were considered in the AMC initiative studied but none were considered in the TCC initiative.

4.3 Merging of Characteristics

Certain characteristics were deemed best to be merged as the respondents were attributing the same meaning to them, even though they were discussed separately in the literature. Some were further refined to better represent the attributed meaning. A summary of the characteristics that were merged after the analysis is presented below.

- C3 [Performance measures being relevant to the business process, the people who are accountable for the process; and the output of the process]. '*Relevant to the business process*' has a similar meaning to C19 - [performance measures integrated with process execution and connected to the KPI and the process steps]. Therefore, C19 was absorbed into C3.
- C9 [All-inclusive/balanced/multi-dimensional set of measures] and C11 [Use of Both Financial, non-financial/Objective, subjective/quantitative, qualitative measures] were discussed by the respondents as having similar meanings.
- C10 [Performance measures should take both long-term and short-term views into account] and C17 [Performance measures should track the past and present performance and performance that influences on future activities/performance] were merged as the 'present' and 'short-term', and the 'future' and 'long-term' were taken by respondents to mean the same.
- C12 [Use of trend and ratio-based performance measures] and C28 [Visuals can make more impact than numbers] are merged because the respondents mentioned that they visualize through ratios and trend lines etc. in graphs.
- C13 [Performance measures should change dynamically and be consistent or coherent with the organizational strategy to support strategy realization] and C14 [Performance measures linked to targets, goals, and objectives] were discussed as meaning the same by respondents.

- C20 [Performance measures consider the organization *as a whole*, to minimize conflict] and C21 [Focus on processes and *integration of functions*]; especially the 'Integration of functions' in C21, was deemed similar to considering 'measures as a whole' in C20.
- C25 [Performance measures should be cost effective to use] was considered a sub part of C24 [Use of automatically collected data and the existing sources of data], as further developments to PPMSs were conducted with the expectation of getting higher benefits from the PPMS in future as per the respondents.
- In C31 [Reports should be made in a simple, *frequent and regular manner*, available constantly for review/use, and not used as a replacement for review meetings]. The aspect of 'Not used as a replacement for review meetings' was not mentioned by respondents and they mentioned that they always review the generated reports. Therefore, C31 was adjusted by adding 'reviewed' to it – [Reports should be made in a simple, frequent and regular manner, available constantly for review/use, and be reviewed regularly].

Further C27 [Maintain *consistency* over time and in *reporting*] and C31 were merged, as 'consistency' in 'reporting' in C27 was deemed similar in meaning by respondents to 'reports should be made in a … frequent and regular manner' in C31.

- C33 […provide comprehensive information to users with easily identifiable and useful relationships…] and C36 […provide information for actionable results. Remedial action …] were merged as respondents did not differentiate between these two ideas. Therefore, C36 was positioned as a subpart of C33.
- C34 [Performance measurement results - Enable consistent benchmarking/comparisons…] and C35 […focus on improvements and inspire and permit employees to monitor, control and further improve …] were merged, as many of the respondents indicated that continuous improvement had been done by benchmarking and through comparisons. Therefore, C34 was positioned as a subpart of C35.

5 PPMS Characteristics Framework and Summary Discussion

With the additions and mergers, 30 PPMS characteristics were derived as the confirmed set of PPMS characteristics. Codes for the confirmed characteristics (*column* 3), codes for the original and merged characteristics (*column* 1) together with the number of literature sources for each of the 38 original characteristics (*column* 2) and sources excluding overlaps related to merged characteristics (*column* 4) and the finalized characteristics (*column* 5) are presented in Table 1.

Table 1. The confirmed 30 PPMS characteristics.

1 Related Old Characteristic(s)	2 Literature Sources before merging	3 New Codes of the confirmed Characteristics (NC)	4 Literature Sources after merging	5 Confirmed Characteristics	6 PMS	7 PPMS	8 PMS & PPMS
C1	11	NC1	11	Clearly defined			X
C2	7	NC2	7	Validity and reliability			X
C3 C19	9 6	NC3	**13**	Relevancy			**X**
C4	23	NC4	23	Simplicity			X
C5	1	NC5	1	Continuous monitoring		X	
C6	7	NC6	7	Brevity	X		
C7	16	NC7	16	Continuous updates			X
C8	1	NC8	1	Use of research	X		
C9 C11	16 28	NC9	**35**	Balanced and multi-dimensional			**X**
C10 C17	6 5	NC10	**11**	Coverage	**X**		
C13 C14	24 17	NC11	**34**	Aligned to strategy and targets			**X**
C15	8	NC12	8	Linked to critical success factors	X		
C16	17	NC13	17	Consider stakeholders			X
C18	7	NC14	7	Linked to rewards	X		
C20 C21	6 11	NC15	**16**	Organization as a whole			**X**
C22	13	NC16	13	Control			X
C23	17	NC17	17	Consistent application			X
C24 C25	6 4	NC18	**10**	Use of existing information sources			**X**

1	2	3	4	5	6	7	8
Related Old Characteristic(s)	Literature Sources before merging	New Codes of the confirmed Characteristics (NC)	Literature Sources after merging	Confirmed Characteristics	PMS	PPMS	PMS & PPMS
C26	4	NC19	4	Updates and resources			X
C27 C31	3 4	NC20	**6**	Reporting and measurement consistency			**X**
C28 C12	4 7	**NC21**	**9**	Visualizing			**X**
C29	11	NC22	11	Fast and timely feedback	X		
C30	9	NC23	9	Accurate feedback			X
C32	10	NC24	10	Mapping contributions			X
C33 C36	11 6	NC25	**15**	Comprehensive and informative			**X**
C34 C35	10 18	NC26	**24**	Focused on improvements			**X**
C37	12	NC27	12	Decision making			X
C38	4	NC28	4	Positive use			X
Newly added		NC29	Not applicable (NA)	Challenge to encourage	NA	NA	NA
Newly added		NC30	NA	Guidance	NA	NA	NA

Using the literature review and case data as the evidence base, these 30 PPMS characteristics were inductively analyzed (through multiple iterations supported by multi-coder corroboration sessions) to identify higher-level themes in which to group them. This resulted in 7 themes, summarized in Fig. 1:

1. Quality characteristics of the PPMS: the designers of PPMSs should take these quality characteristics into account as they influence efficiency and effectiveness, to assure that the PPMS is of high quality.

2. Measurement scope of the PPMS: are the characteristics that ensure the provision of an overall assessment of the performance of the processes within an organization. The measurements should include financial and non-financial measures, as well as short and long term measures.
3. Contextual features considered in designing the PPMS: refers to characteristics relating to critical success factors, key business drivers, the organizational strategy, targets and goals, and the stakeholder needs.
4. Relationship to organizational systems and structures: characteristics regarding the importance of linking PPMSs to other systems and structures in the organization such as linking PPMS to rewards systems.
5. Efficiency of information gathering and use: characteristics that stress the need for PPMSs to be cost effective while using the existing sources of data and formally training the employees in the use of PPMS.
6. Feedback and reporting: characteristics of how the PPMS should conduct reporting and provide feedback. Consistency and transparency of generated information and frequent reporting cycles are emphasized.
7. Potential uses of the feedback generated through the PPMS: The list is not exhaustive but shows the characteristics referring to potential use cases of PPMSs.

The PPMS Characteristics (NC1–NC30 – see Table 1 for further supporting detail) mapped against these seven themes are presented in Fig. 1, forming the final PPMS Characteristics framework. Even though some characteristics were mentioned only in relation to PMS (e.g. C6, C8 etc.) or PPMS (C5), most of the others were mentioned related to both PMS and PPMS in literature (these details are depicted in the last three columns in Table 1). But based on the respondents of the two case studies, we argue that all 30 characteristics should be embedded into a PPMS as they serve different purposes (indicated by the names of the themes into which they are grouped).

Measuring process performance is a precondition for analyzing and subsequently improving business processes [51], and process performance measurement is a critical phase of any process improvement lifecycle [52]. As described by Hernaus, [11] significant benefits can be gained in strategic and operational management through the use of effective process performance measures. However, having the 'right' measurement system and the supporting process-centric data has been a long-term challenge in the field [51], due to the lack of PPMSs. This lack of effective measurement systems and accurate and easily obtainable process data (past and present) hinders process improvement initiatives [53, 54], the ability to accurately identify pain points, and benchmark or show the impact of process improvements efforts.

The study results presented here provide the necessary evidence-based guidelines to address this gap of developing a PPMS, in order to measure process performance data by identifying the characteristics necessary for a PPMS. These characteristics become a useful checklist for practitioners designing and/or evaluating PPMSs. While recent developments in Business Process Management Systems (BPMSs), especially data-centric enhancements, have improved access to process-centric data, more holistic guidance on how such access to process-centric data should be designed and integrated within organizational contexts is still lacking. This empirically confirmed framework with the seven themes and 30 characteristics provides a holistic overview, complimenting such technical developments in BPMSs; it points to how process performance measurement should contribute towards quality assurance, cost management, integration to different systems and structures, and how process performance measures need to dynamically evolve within diverse contexts. It also captures how the feedback loops and reporting resulting from PPMS outcomes should be set up and how to best operationalize the insights obtained from deploying a PPMS. This framework guides organizations on how to move away from current '*ad-hoc*' [13] process measurement efforts, reduce existing uncertainties [15], and (re-) design PPMSs appropriately [14] with a holistic and long-term view.

6 Conclusion

Conceptualizing PPMSs is a critical gap that is unaddressed, and this study attempts to fill that gap by presenting an empirically supported PPMS characteristics framework that identifies what characteristics a PPMS should contain. A multi-staged study design was followed where 38 PPMS characteristics identified through a structured literature review were re-specified and confirmed with two in-depth case studies resulting in 30 PPMS characteristics, which were then categorized into 7 themes. This empirically confirmed PPMS framework (the first of its kind) can serve as a useful reference point for PPMS practice and future research. Compared to the available studies on PMS/PPMS characteristics (within the identified fifty-six papers) this is a complete collection of characteristics synthesized to make the scattered knowledge on characteristics to a single place. The list of characteristics can be used as a complete check list for practitioners to design PPMSs and to review and adapt traditional PMSs. The collection and synthesis were done in a rigorous manner. The method followed in building the themes as well as the characteristics is tracked through Excel spread sheets and Nvivo data management tool. Thus, having a trail of evidence between initial extracts from the papers and meta themes. The identified list facilitates in making sense and have eased the attempts of taking actions on PPMS. Themes provide a clearer simplified and actionable set of areas that needs to be addressed when designing PPMSs or reviewing or adapting traditional PMSs to suit process-centric settings.

While a rigorous literature and case study-based approach was followed, the results may have some limitations. While a literature based a priori models provide guidance for empirical work, they can also influence 'what is sought for' in the empirical settings; hence introduce some bias. The framework may also be affected by usual limitations of qualitative research such as; selection- and researcher-bias, despite the protocols followed to minimize these. Furthermore, we acknowledge that the PPMS characteristics can be dynamic in nature (i.e. evolve and change in diverse contexts and over time). Given that the cases studied here were from the manufacturing sector, we recommend future research that investigates and validates the PPMS characteristics framework in other sectors. Future research can also focus on identifying the PPMS characteristics that are differently relevant to diverse process improvement project types [49]; and better understand which characteristics are essential for all project types and which are important for only some, as well as how they change in diverse contexts and over time. Future research can also investigate evolving PPMS design-steps (i.e.

Blasini et al. 2018), in particular to better understand when and how the identified PPMS characteristics should be integrated into the PPMS design process.

Appendix A

PPMS characteristics extracted from the Structured Literature Review, available at; http://www.workflowpatterns.com/janitha/Appendix%20A.pdf

Appendix B

Case study evidence supporting the changes made to the a priori PPMS Characteristics, available at; http://www.workflowpatterns.com/janitha/Appendix%20B.pdf

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